

AI Project 2: Data Classification Using AI

DecodeLabs Industrial Training Kit — Batch 2026

Project objective: Build a basic classification model using a small dataset. The supplied project brief requires loading and understanding data, splitting it into training/testing sets, and applying a simple classification algorithm. ■filecite■turn0file0■L33-L46■

1. Dataset

The Iris benchmark contains 150 samples, 3 classes, and 4 feature dimensions. The four features used are sepal length, sepal width, petal length, and petal width. The three classes are setosa, versicolor, and virginica.

2. Data Preparation

The dataset is shuffled and divided using an 80/20 train-test split. StandardScaler is fitted only on the training data and then applied to both training and testing data. This follows the supplied blueprint's emphasis on scaling and randomized splitting to reduce order bias. ■filecite■turn1file0■L22-L31■

3. Classification Algorithm

The classifier is K-Nearest Neighbors (KNN) with $k=5$. KNN predicts a new sample from the majority class among its nearest neighbors. The supplied project visual specifically presents $K=5$ and the proximity principle. ■filecite■turn1file0■L33-L37■

4. Implementation Workflow

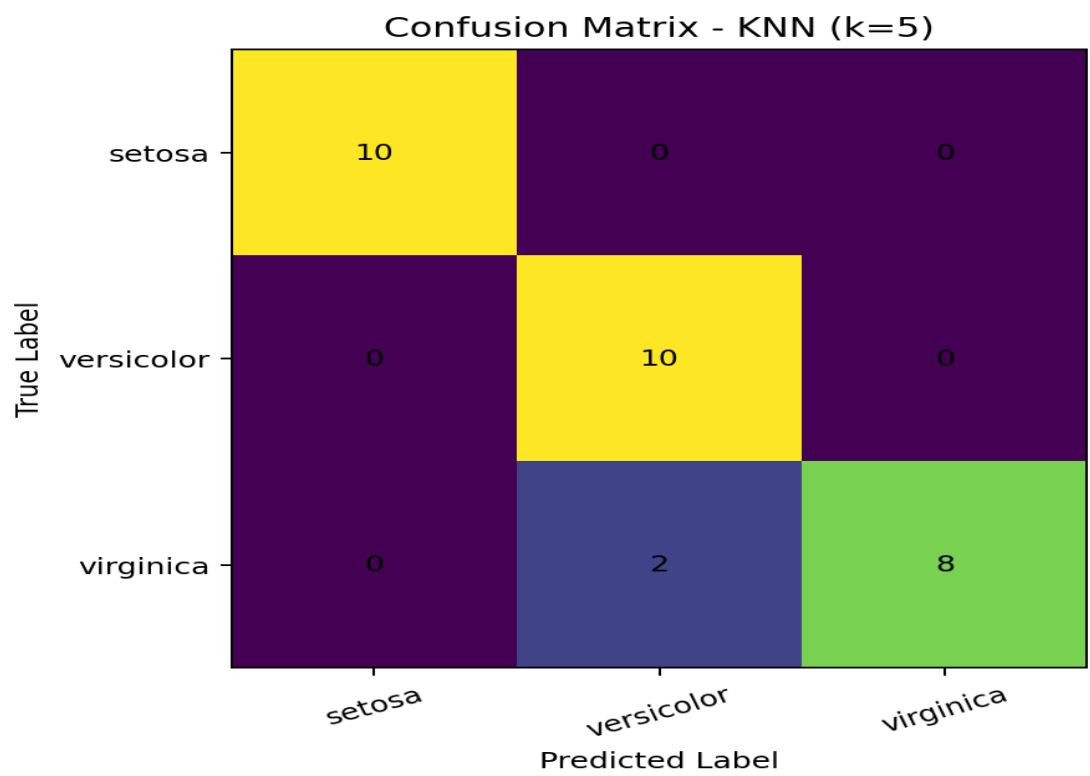
The implementation follows the project pipeline: instantiate the KNN model, fit it on the training data, and predict labels for the test data. ■filecite■turn1file0■L39-L40■

5. Evaluation

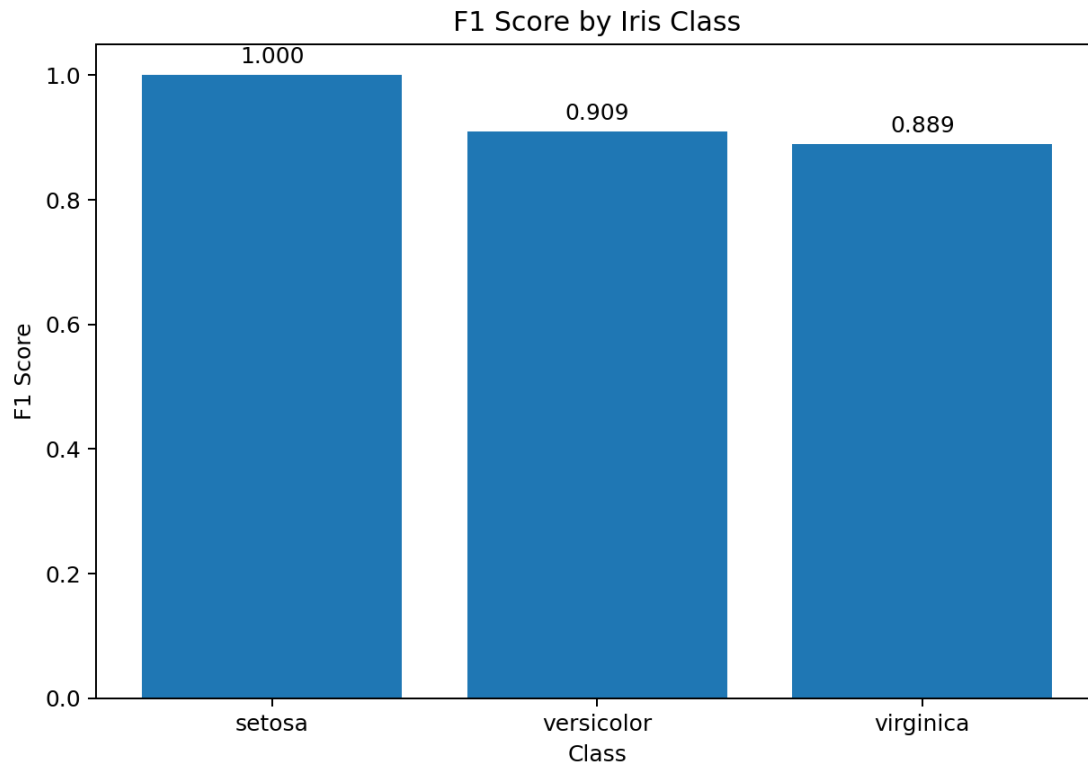
The model achieved an accuracy of 93.33% and a weighted F1 score of 0.9327 on the held-out test set. The project brief emphasizes that accuracy should not be the only diagnostic for classification; it introduces the confusion matrix and F1 score as deeper evaluation measures. ■filecite■turn1file0■L41-L43■

Metric	Value
Accuracy	93.33%
Weighted Precision	0.9444
Weighted Recall	0.9333
Weighted F1 Score	0.9327

6. Confusion Matrix



7. F1 Score by Class



8. Conclusion

This project demonstrates the supervised-learning workflow requested in Project 2: data loading, preparation, train-test splitting, scaling, KNN model training, prediction, and evaluation. It provides a complete beginner-friendly classification pipeline that can be run locally and uploaded to GitHub as a portfolio project.